Feature Best / Worst Case Reference

Geospatial Energy Profiling Pipeline · 512 m × 512 m bbox · 6 sources · ~33 features · Best = highest energy-relevant signal · Worst = lowest / absent signal

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SOURCE 1 — ESA WorldCover (10m land cover)

wc_dominant_pct

BEST CASE	WORST CASE
Amazon rainforest interior (-3.58, 23.52) **Value:** ~100% Pure unbroken tree_cover with zero edge effects across the full 512 m bbox. Open satellite view	**Lagos peri-urban fringe** (6.6, 3.35) **Value:** ~28–32% Explosive informal growth mixes built_up, bare soil, water, and vegetation with no single class dominating. Open satellite view

wc_class_count

BEST CASE	WORST CASE
Rotterdam port (51.93, 4.14) **Value:** 6–7 classes Industrial port combines water, built_up, bare soil, wetland, cropland fringe, and transport corridors within 512 m. Open satellite view	**Sahara sand sea, Algeria** (26.0, 3.0) **Value:** 1 class Endless bare_sparse erg with zero vegetation or water intrusion. Open satellite view

wc_std

BEST CASE	WORST CASE
Amazon riverbank, Brazil (-3.06, -59.83) **Value:** ~0.45+ Sharp water/forest boundary maximises pixel-code variance within the bbox. Open satellite view	**Iowa corn belt** (41.87, -93.1) **Value:** ~0.0 100 % cropland monoculture; every pixel the same class code so std collapses to zero. Open satellite view

wc_energy_score_mean

BEST CASE	WORST CASE
Rotterdam port (51.93, 4.14) **Value:** ~2.8–3.0 Dense built_up (score 3.0) dominates the entire bbox with industrial infrastructure. Open satellite view	**Antarctica interior** (-75.0, 0.0) **Value:** ~0.1 Permanent snow_ice (score 0.1) across every pixel; no energy-relevant land use. Open satellite view

SOURCE 2 — GHSL (Global Human Settlement Layer)

ghsl_built_surface_m2

BEST CASE	WORST CASE
Manhattan midtown (40.755, -73.984) **Value:** ~7,500–8,000 m²/px Among the densest built footprints on Earth; nearly every 100 m pixel fully covered. Open satellite view	**Sahara desert, Algeria** (26.0, 3.0) **Value:** 0 m² No human settlement; GHSL built-surface layer returns zero. Open satellite view

ghsl_building_height_m

BEST CASE	WORST CASE
Hong Kong Central (22.282, 114.158) **Value:** ~20–25 m mean AGBH Hyper-dense skyscraper district with some of the highest average heights in GHSL. Open satellite view	**Rural Kansas** (38.85, -96.6) **Value:** 0 m Open farmland with no structures; GHSL AGBH returns zero. Open satellite view

ghsl_population_per_km2

BEST CASE	WORST CASE
Dhaka old city (23.723, 90.407) **Value:** ~80,000–110,000/km² One of the highest recorded population densities on the planet. Open satellite view	**Greenland ice sheet** (72.0, -40.0) **Value:** 0 persons/km² Uninhabited permanent ice; GHSL population raster returns zero. Open satellite view

ghsl_degurba_class

BEST CASE	WORST CASE
Tokyo Shinjuku (35.689, 139.692) **Value:** dense_urban_centre Meets all GHSL thresholds for contiguous high-density urban fabric. Open satellite view	**Norwegian fjord hinterland** (61.0, 7.0) **Value:** rural Isolated valley hamlet far below any urban cluster density threshold. Open satellite view

SOURCE 3 — Global Solar Atlas (World Bank / Solargis)

solar_pvout_kwh_kwp_day

BEST CASE	WORST CASE
Atacama Desert, Chile (-24.5, -69.5) **Value:** ~6.2–6.5 kWh/kWp/day Extreme altitude, zero cloud cover, intense irradiance year-round gives the global PVOUT maximum. Open satellite view	**Tromsø, Norway** (69.65, 18.95) **Value:** ~1.0–1.2 kWh/kWp/day Months of polar night drag the annual mean to near-minimum globally. Open satellite view

solar_ghi_kwh_m2_day

BEST CASE	WORST CASE
Saharan interior, Libya (27.0, 14.0) **Value:** ~7.2–7.5 kWh/m²/day Cloudless desert with near-perpendicular solar incidence; among the top GHI pixels globally. Open satellite view	**Reykjavik, Iceland** (64.13, -21.93) **Value:** ~1.6–1.8 kWh/m²/day High latitude, persistent overcast, and short winter days minimise annual mean GHI. Open satellite view

solar_dni_kwh_m2_day

BEST CASE	WORST CASE
Atacama Desert, Chile (-24.5, -69.5) **Value:** ~8.5–9.0 kWh/m²/day Thin dry atmosphere at 2,400 m altitude maximises direct beam fraction; CSP plants operate here for this reason. Open satellite view	**Atlantic coast of Ireland** (53.0, -8.0) **Value:** ~1.0–1.3 kWh/m²/day Persistent maritime cloud cover scatters most solar radiation, collapsing the direct-normal component. Open satellite view

SOURCE 4 — ERA5 (Copernicus CDS)

era5_ssrd_j_m2_day

BEST CASE	WORST CASE
BEST CASE **Saharan interior, Libya** (27.0, 14.0) **Value:** ~26–28 MJ/m²/day Cloudless desert atmosphere allows near-maximum downward shortwave flux at the surface. Open satellite view	**WORST CASE** **Antarctica interior (winter)** (-75.0, 0.0) **Value:** ~0 MJ/m²/day Polar night means zero solar input; ERA5 SSRD drops to essentially null for months. Open satellite view

SOURCE 5 — VIIRS / MODIS via Google Earth Engine

viirs_ntl_nw_cm2_sr

BEST CASE	WORST CASE
BEST CASE **Las Vegas Strip** (36.114, -115.172) **Value:** ~180–220 nW/cm²/sr Highest sustained artificial light density in North America; casino lighting saturates the sensor. Open satellite view	**WORST CASE** **Congo Basin interior** (-0.58, 23.52) **Value:** ~0.05–0.15 nW/cm²/sr Vast uninhabited rainforest with no electrification; one of the darkest land surfaces on the VIIRS map. Open satellite view

viirs_ndvi

BEST CASE	WORST CASE
BEST CASE **Amazon rainforest, Brazil** (-3.58, -60.02) **Value:** ~0.82–0.88 Year-round closed-canopy tropical forest maximises the NIR/red ratio NDVI measures. Open satellite view	**WORST CASE** **Rub al Khali, Saudi Arabia** (21.0, 51.0) **Value:** ~0.02–0.05 Hyperarid sand desert with zero photosynthetic activity; NDVI near the theoretical minimum. Open satellite view

viirs_surface_refl

BEST CASE	WORST CASE
BEST CASE **Chott el Djerid, Tunisia** (33.7, 8.4) **Value:** ~0.55–0.65 White salt crust acts as a near-perfect red-band reflector; one of the brightest land surfaces globally. Open satellite view	**WORST CASE** **Congo River water surface** (-0.06, 18.26) **Value:** ~0.01–0.03 Dark turbid water absorbs nearly all red-band radiation; among the lowest MODIS red values for inland water. Open satellite view

SOURCE 6 — OpenStreetMap (Overpass API)

osm_building_count

BEST CASE	WORST CASE
BEST CASE **Central Amsterdam** (52.372, 4.895) **Value:** 400–500 buildings Dense 17th-century canal ring with narrow plots; one of the highest OSM building densities in Europe. Open satellite view	**WORST CASE** **Mongolian steppe** (47.5, 106.0) **Value:** 0 Open grassland with no permanent structures; OSM returns an empty building set. Open satellite view

osm_building_type

BEST CASE	WORST CASE
Dhaka old city (apartments) (23.723, 90.407) **Value:** apartments Overwhelming residential density means apartment tags dominate the building way types. Open satellite view	**Rotterdam port (industrial)** (51.93, 4.14) **Value:** industrial/warehouse Port logistics zone mapped almost exclusively with industrial and warehouse building tags. Open satellite view

osm_building_height_mean

BEST CASE	WORST CASE
Chicago Loop (41.882, -87.629) **Value:** ~70–90 m Dense skyscraper core where OSM contributors have diligently tagged heights; Willis Tower anchors the mean. Open satellite view	**Rural northern Vietnam** (20.0, 106.0) **Value:** ~3–4 m Single-storey tile-roofed houses with near-zero height tags in OSM. Open satellite view

osm_building_material

BEST CASE	WORST CASE
Ruhr region, Germany (51.45, 6.95) **Value:** brick / concrete Industrial heritage and post-war reconstruction are tagged with building:material in OSM's well-mapped German corpus. Open satellite view	**Mongolian steppe** (47.5, 106.0) **Value:** None No mapped buildings; material attribute absent. Open satellite view

osm_building_start_date

BEST CASE	WORST CASE
City of London (historic core) (51.508, -0.076) **Value:** e.g. 1677 OSM contributors have tagged start_date on historic City buildings; earliest dates reach back centuries. Open satellite view	**Mongolian steppe** (47.5, 106.0) **Value:** None No buildings mapped; start_date tag absent. Open satellite view

osm_roof_shape

BEST CASE	WORST CASE
Central Amsterdam (52.372, 4.895) **Value:** gabled Dutch canal houses are extensively tagged with roof:shape=gabled by OSM contributors. Open satellite view	**Mongolian steppe** (47.5, 106.0) **Value:** None No buildings mapped; roof attributes absent. Open satellite view

osm_roof_material

BEST CASE	WORST CASE
Central Amsterdam (52.372, 4.895) **Value:** tiles Amsterdam's historic core has highly detailed OSM data including ceramic roof tile material tags. Open satellite view	**Sahara desert, Algeria** (26.0, 3.0) **Value:** None No structures present; roof material tag cannot exist. Open satellite view

osm_power_substation

BEST CASE	WORST CASE
Rotterdam port (51.93, 4.14) **Value:** True Heavy industrial electricity demand means substations are present and well-mapped. Open satellite view	**Sahara desert, Algeria** (26.0, 3.0) **Value:** None No power grid infrastructure mapped or present in uninhabited desert. Open satellite view

osm_generator_source (solar)

BEST CASE	WORST CASE
Ouarzazate Solar Complex, Morocco (30.93, -6.91) **Value:** solar Noor I/II/III CSP and adjacent PV fields are explicitly mapped with generator:source=solar. Open satellite view	**Amazon rainforest interior** (-3.06, -59.83) **Value:** None Dense canopy, zero grid infrastructure; no solar generators mapped or physically present. Open satellite view

osm_generator_source (wind)

BEST CASE	WORST CASE
Hornsea offshore wind, UK (53.9, 1.9) **Value:** wind World's largest offshore wind farm cluster; turbines mapped with generator:source=wind. Open satellite view	**Manhattan midtown** (40.755, -73.984) **Value:** None Dense urban core with no space or permission for wind turbines; tag absent. Open satellite view

osm_landuse

BEST CASE	WORST CASE
Ruhr industrial region, Germany (51.45, 6.95) **Value:** industrial Heart of Europe's largest industrial conurbation; landuse=industrial tags are dense and extensive. Open satellite view	**Black Forest, Germany** (48.0, 8.2) **Value:** forest (no industrial) Protected woodland with landuse=forest; no industrial tag present. Open satellite view

osm_highway

BEST CASE	WORST CASE
Ruhr motorway network, Germany (51.45, 6.95) **Value:** motorway Europe's densest motorway interchange area; multiple Autobahn ways present in any 512 m bbox. Open satellite view	**Mongolian steppe** (47.5, 106.0) **Value:** None Unmapped, unpaved tracks at best; OSM highway tags absent. Open satellite view

osm_surface

BEST CASE	WORST CASE
Köln city centre (50.943, 6.958) **Value:** asphalt Heavily mapped German city where surface=asphalt is tagged on nearly every way. Open satellite view	**Mongolian steppe** (47.5, 106.0) **Value:** None No paved roads; surface tags absent from unmapped dirt tracks. Open satellite view

osm_railway

BEST CASE **Köln Hauptbahnhof** (50.943, 6.958) **Value:** True Major rail hub with 12+ tracks visible; railway ways densely mapped by OSM community. Open satellite view	**WORST CASE** **Mongolian steppe** (47.5, 106.0) **Value:** None Hundreds of kilometres from any rail line; OSM returns no railway ways. Open satellite view

osm_pipeline

BEST CASE **Tengiz oil field, Kazakhstan** (45.47, 53.12) **Value:** True Major crude export terminal with multiple mapped pipeline ways radiating from the field. Open satellite view	**WORST CASE** **Black Forest, Germany** (48.0, 8.2) **Value:** None Protected forest reserve with no pipeline infrastructure mapped. Open satellite view

osm_amenity (charging_station)

BEST CASE **Zurich motorway hub, Switzerland** (47.45, 8.56) **Value:** charging_station Major highway interchange with multiple Tesla Supercharger and CCS stations mapped in OSM. Open satellite view	**WORST CASE** **Lagos peri-urban fringe** (6.6, 3.35) **Value:** None EV infrastructure essentially absent in informal peri-urban West Africa. Open satellite view

osm_supermarket

BEST CASE **Central Amsterdam** (52.372, 4.895) **Value:** True Dense mixed-use urban core with Albert Heijn and other chains mapped at high density. Open satellite view	**WORST CASE** **Mongolian steppe** (47.5, 106.0) **Value:** None No retail infrastructure mapped or present; nearest settlement tens of kilometres away. Open satellite view